THE FOURTH HAT IN PROJECTIVE COORDINATES

1. The construction in coordinates

1.1. **Notation.** Let q be a prime power and $\mathbb{F} = \mathrm{GF}(q)$, $\mathbb{F}^* = \mathbb{F} \setminus \{0\}$. A point of $\mathrm{PG}(2,q)$ is written $(x{:}y{:}z)$ and a line $[u{:}v{:}w]$, with $(x,y,z),(u,v,w) \in \mathbb{F}^3 \setminus \{0\}$ up to nonzero scalars; the point $(x{:}y{:}z)$ is on the line $[u{:}v{:}w]$ if and only if $ux + vy + wz = 0$. We fix

$$\ell = [1{:}0{:}0] \text{ (the line } x = 0), \qquad A = (1{:}0{:}0), \qquad A \notin \ell,$$

and name the points of ℓ and the lines through A:

$$X_t = (0{:}t{:}1)\ (t \in \mathbb{F}), \quad P = (0{:}1{:}0); \qquad \lambda_t = [0{:}1{:}{-}t]\ (t \in \mathbb{F}), \quad \lambda_\infty = [0{:}0{:}1].$$

Every point and every line has exactly one representative in the following lists.

points	number	lines	number
$A = (1{:}0{:}0)$	1	$\ell = [1{:}0{:}0]$	1
$X_t = (0{:}t{:}1)$, $t \in \mathbb{F}$	q	$\lambda_t = [0{:}1{:}{-}t]$, $t \in \mathbb{F}$	q
$P = (0{:}1{:}0)$	1	$\lambda_\infty = [0{:}0{:}1]$	1
$(x{:}y{:}1)$, $x \in \mathbb{F}^*$, $y \in \mathbb{F}$	$q(q-1)$	$[u{:}1{:}w]$, $u \in \mathbb{F}^*$, $w \in \mathbb{F}$	$q(q-1)$
$(x{:}1{:}0)$, $x \in \mathbb{F}^*$	$q-1$	$[1{:}0{:}w]$, $w \in \mathbb{F}^*$	$q-1$

The points not on ℓ are those with $x \neq 0$; the lines not through A are those with $u \neq 0$. We use the following facts, each a one-line computation.

(F1) $\lambda_t \cap \ell = X_t$ and $\lambda_\infty \cap \ell = P$. The points of λ_t are A and $(x{:}t{:}1)$, $x \in \mathbb{F}$ (with X_t at $x = 0$); the points of λ_∞ are A and $(x{:}1{:}0)$, $x \in \mathbb{F}$ (with P at $x = 0$). In particular a point $(x{:}y{:}1)$ with $x \neq 0$ lies on λ_y, and a point $(x{:}1{:}0)$ lies on λ_∞.

(F2) The lines through X_t are ℓ and $[w{:}1{:}{-}t]$, $w \in \mathbb{F}$ (with λ_t at $w = 0$). The lines through P are ℓ, λ_∞ and $[1{:}0{:}w]$, $w \in \mathbb{F}^*$.

(F3) If $u = (x{:}y{:}z)$ with $x \neq 0$ and $s \in \mathbb{F}$, then u and X_s lie on $[(sz - y)/x{:}1{:}{-}s]$, which is not ℓ.

(F4) A line $L = [u{:}v{:}w]$ with $u \neq 0$ meets λ_s in $(-(vs + w)/u{:}s{:}1)$, which is not A; and $[u{:}1{:}w]$ (with $u \neq 0$) meets ℓ in X_{-w}.

A correlation. Let δ map the point $(x{:}y{:}z)$ to the line $[x{:}z{:}{-}y]$ and the line $[u{:}v{:}w]$ to the point $(u{:}w{:}{-}v)$. Since $ux + wz + (-v)(-y) = ux + vy + wz$, δ maps incident pairs to incident pairs; it is a bijection from points to lines and from lines to points, and

$$\delta(A) = \ell, \quad \delta(\ell) = A, \quad \delta(X_t) = \lambda_t, \quad \delta(\lambda_t) = X_t, \quad \delta(P) = \lambda_\infty, \quad \delta(\lambda_\infty) = P.$$

Moreover $(x{:}y{:}z)$ lies on $\delta(x{:}y{:}z)$ if and only if $x^2 + zy - yz = x^2 = 0$, that is, if and only if the point lies on ℓ. We use δ only for the final matching (Step (C7)) and in Remark 7; the proofs below do not depend on it.

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Dictionary. In the affine coordinates of [1], $(x,y) = (x{:}y{:}1)$, $P_m = (1{:}m{:}0)$, $P = (0{:}1{:}0)$, $[m,b] = [m{:}{-1}{:}b]$ (the line $y = mx + b$), $L_c = [1{:}0{:}{-c}]$ (the line $x = c$) and $L = [0{:}0{:}1]$. Hence

$$L_0 = \ell, \quad P_0 = A, \quad (0,s) = X_s, \quad [0,s] = \lambda_s, \quad L = \lambda_\infty,$$

and the point P is the same. In the coordinate-free description of the four-hat construction, $AX_s = \lambda_s$, $AP = \lambda_\infty$, and $\ell^* = \{P\} \cup \{X_t : t \notin \{0,1,c\}\}$.

1.2. **The construction of Araujo-Pardo and López.** Let $q \geq 3$, fix $a \in \mathbb{F} \setminus \{0,1\}$ and put $S_{\mathrm{AL}} = \{0,1,a\}$. Add the points Q, H_0, H_1, H_a and the lines M, K_0, K_1, K_a (in [1]: Q_0, $\widehat{(0,s)}$, M_0, $\widehat{[0,s]}$). The structure $\mathcal{A}$ has these incidences:

(A1) every incidence of $\mathrm{PG}(2,q)$ except $X_a \in \lambda_a$;
(A2) H_s ($s \in S_{\mathrm{AL}}$) lies on $[w{:}1{:}{-s}]$ for every $w \in \mathbb{F}$;
(A3) K_s ($s \in S_{\mathrm{AL}}$) contains $(x{:}s{:}1)$ for every $x \in \mathbb{F}$;
(A4) M contains P, X_t for $t \in \mathbb{F} \setminus \{0,1\}$, and H_0, H_1, H_a;
(A5) Q lies on λ_∞, on λ_t for $t \in \mathbb{F} \setminus \{0,1\}$, and on K_0, K_1, K_a;
(A6) $A \in K_a$, $H_a \in \ell$, $H_0 \in K_1$, $H_1 \in K_0$.

By (F2), H_s lies on the q lines through X_s other than ℓ; by (F1), K_s contains the q points of λ_s other than A. In the notation of [1], (A2) is $\widehat{(0,s)} \sim [t,s]$ (as $[t,s] = [{-t}{:}1{:}{-s}]$) and (A3) is $\widehat{[0,s]} \sim (t,s)$.

Neighbour sets. The new vertices of $\mathcal{A}$ have these neighbours, each $q+2$ of them:

Q:	λ_∞, λ_t ($t \in \mathbb{F} \setminus \{0,1\}$), K_0, K_1, K_a	M:	P, X_t ($t \in \mathbb{F} \setminus \{0,1\}$), H_0, H_1, H_a
H_0:	$[w{:}1{:}0]$ ($w \in \mathbb{F}$), M, K_1	K_0:	$(x{:}0{:}1)$ ($x \in \mathbb{F}$), Q, H_1
H_1:	$[w{:}1{:}{-1}]$ ($w \in \mathbb{F}$), M, K_0	K_1:	$(x{:}1{:}1)$ ($x \in \mathbb{F}$), Q, H_0
H_a:	$[w{:}1{:}{-a}]$ ($w \in \mathbb{F}$), M, ℓ	K_a:	$(x{:}a{:}1)$ ($x \in \mathbb{F}$), Q, A

The vertices of $\mathrm{PG}(2,q)$ whose neighbourhood changes are: A ($+K_a$); P and X_t, $t \notin \{0,1,a\}$ ($+M$); X_a ($-\lambda_a$, $+M$, $+K_a$); X_0, X_1 ($+K_0$, resp. $+K_1$); $(x{:}s{:}1)$, $x \neq 0$, $s \in S_{\mathrm{AL}}$ ($+K_s$); ℓ ($+H_a$); λ_∞ and λ_t, $t \notin \{0,1,a\}$ ($+Q$); λ_a ($-X_a$, $+Q$, $+H_a$); λ_0, λ_1 ($+H_0$, resp. $+H_1$); $[u{:}1{:}{-s}]$, $u \neq 0$, $s \in S_{\mathrm{AL}}$ ($+H_s$). Hence every vertex has degree $q+2$ except the points $(x{:}y{:}1)$ ($x \neq 0$, $y \notin S_{\mathrm{AL}}$) and $(x{:}1{:}0) = P_{1/x}$ ($x \neq 0$), and the lines $[x{:}{-1}{:}y] = [x,y]$ ($x \neq 0$, $y \notin S_{\mathrm{AL}}$) and $[1{:}0{:}{-i}] = L_i$ ($i \neq 0$), which have degree $q+1$; there are $(q-1)(q-2)$ of each. The matching of [1] is

$$(x,y) \sim [x,y] \ \ (x \neq 0,\ y \notin S_{\mathrm{AL}}), \qquad P_i \sim L_i,$$

non-incident pairs because $(x,y) \in [x,y]$ iff $x^2 = 0$ and $P_i \notin L_i$ always. In [1] the second family is indexed by $i \in \mathbb{F} \setminus \{0,1,a\}$; then P_1, L_1, P_a, L_a keep degree $q+1$. With $i \in \mathbb{F}^*$ the graph is $(q+2)$-regular of order $2(q^2+q+5)$. The map δ of §1.1 gives another such matching, $p \sim \delta(p)$, as in Step (C7) below. (The maximum degree and the diameter do not depend on the matching.)

Diameter of $\mathcal{A}$. The proofs of Lemmas 3 and 4 below apply to $\mathcal{A}$ with these changes: in (P1) and (L1) only $s = a$ occurs; the pairs $\{H_0, X_1\}$, $\{H_1, X_0\}$, $\{Q, H_0\}$, $\{Q, H_1\}$ use K_1, K_0, K_1, K_0; the pairs $\{K_0, \lambda_1\}$, $\{K_1, \lambda_0\}$ use H_1, H_0; the pairs $\{M, K_0\}$, $\{M, K_1\}$ use H_1, H_0. All other cases are word for word the same with T replaced by $\{0,1\}$. So $\mathcal{A}$ also has diameter 3, for every prime power $q \geq 3$.

1.3. **The four-hat construction.** Let $q \geq 4$ and fix distinct $a, c \in \mathbb{F} \setminus \{0,1\}$ (this needs $q \geq 4$). Put

$$S = \{0,1,a,c\}, \qquad T = \{0,1,c\}, \qquad \sigma\colon\ 0 \mapsto c \mapsto 1 \mapsto 0 \quad (\text{a 3-cycle on } T).$$

Add five points Q, H_0, H_1, H_a, H_c and five lines M, K_0, K_1, K_a, K_c. The structure $\mathcal{S} = \mathcal{S}(q; a, c)$ has exactly the following incidences.

(C1) Every incidence of $\mathrm{PG}(2,q)$ except the four incidences $X_s \in \lambda_s$, $s \in S$; and the three new incidences $X_t \in \lambda_{\sigma(t)}$, $t \in T$:

$$X_0 \in \lambda_c, \qquad X_c \in \lambda_1, \qquad X_1 \in \lambda_0.$$

(C2) For $s \in S$, H_s lies on $[w{:}1{:}{-}s]$ for every $w \in \mathbb{F}$.
(C3) For $s \in S$, K_s contains $(x{:}s{:}1)$ for every $x \in \mathbb{F}$.
(C4) M contains P, X_t for every $t \in \mathbb{F} \setminus T$, and H_0, H_1, H_a, H_c.
(C5) Q lies on λ_∞, on λ_t for every $t \in \mathbb{F} \setminus T$, and on K_0, K_1, K_a, K_c.
(C6) $A \in K_a$, $H_a \in \ell$, and $H_t \in K_{\sigma(t)}$ for $t \in T$:

$$H_0 \in K_c, \qquad H_c \in K_1, \qquad H_1 \in K_0.$$

(C7) (Optional filler.) Join each point p in $\{(x{:}y{:}1) : x \neq 0,\ y \notin S\} \cup \{(x{:}1{:}0) : x \neq 0\}$ to the line $\delta(p)$:

$$(x{:}y{:}1) \sim [x{:}1{:}{-}y] \quad (x \neq 0,\ y \notin S), \qquad (x{:}1{:}0) \sim [x{:}0{:}{-}1] = [1{:}0{:}{-}1/x] \quad (x \neq 0).$$

Let $G = G(q; a, c)$ be the incidence graph of $\mathcal{S}$ (steps (C1)–(C6)) and $R = R(q; a, c)$ the graph with the filler (C7) added. Both are bipartite with $2(q^2 + q + 1 + 5) = 2(q^2 + q + 6)$ vertices.

The rotation as a map. By (F1), in $\mathrm{PG}(2,q)$ the incidences between the points of ℓ and the lines through A form the perfect matching $X_t \leftrightarrow \lambda_t$, $P \leftrightarrow \lambda_\infty$. Let $\pi\colon \mathbb{F} \to \mathbb{F}$ be σ on T and the identity on $\mathbb{F} \setminus T$. Step (C1) replaces this matching by $X_t \leftrightarrow \lambda_{\pi(t)}$ ($t \in \mathbb{F} \setminus \{a\}$), $P \leftrightarrow \lambda_\infty$, and leaves X_a and λ_a unmatched. All other incidences of $\mathrm{PG}(2,q)$ are kept. (In $\mathcal{A}$, π is the identity and again only $X_a \leftrightarrow \lambda_a$ is deleted.)

Changes relative to $\mathcal{A}$. Take $\mathcal{A}$ with the same a and choose $c \in \mathbb{F} \setminus \{0, 1, a\}$. Then $\mathcal{S}$ is obtained from $\mathcal{A}$ by exactly these changes:

(Δ1) add the point H_c, on the q lines $[w{:}1{:}{-}c]$ ($w \in \mathbb{F}$), on M and on K_1; add the line K_c, through the q points $(x{:}c{:}1)$ ($x \in \mathbb{F}$), through Q and through H_0;
(Δ2) remove $X_0 \in \lambda_0$, $X_1 \in \lambda_1$, $X_c \in \lambda_c$, and add $X_0 \in \lambda_c$, $X_c \in \lambda_1$, $X_1 \in \lambda_0$;
(Δ3) remove $X_c \in M$ and $Q \in \lambda_c$;
(Δ4) remove $H_0 \in K_1$ (it is replaced by $H_0 \in K_c$ and $H_c \in K_1$ from the first item; $H_1 \in K_0$ is kept).

The core gains $2(q+2)$ incidences at H_c and K_c (they are not incident with each other) and three from the rotation, and loses six; that is, $2q+1$ more incidences than $\mathcal{A}$. If both structures use the filler $p \sim \delta(p)$, the fillers differ only in that the points $(x{:}c{:}1)$ and the lines $[x{:}1{:}{-}c]$, $x \neq 0$, are no longer matched (they now meet K_c and H_c).

1.4. **Degrees.**

Lemma 1 (Degrees)**.** *Table 1 lists the neighbourhood of every vertex of G. In particular G has maximum degree $q+2$, and its vertices of degree $q+1$ are exactly the points $(x{:}y{:}1)$ ($x \neq 0$, $y \notin S$) and $(x{:}1{:}0)$ ($x \neq 0$), and the lines $[u{:}1{:}w]$ ($u \neq 0$, $-w \notin S$) and $[1{:}0{:}w]$ ($w \neq 0$); there are $(q-1)(q-3)$ of each. The graph R is $(q+2)$-regular.*

Proof. Each row is (F1) or (F2) (the plane neighbours) corrected by (C1)–(C6). We check the rows one class at a time; in each case the listed neighbours are distinct, so the degree is their number.

Points. A: the $q+1$ lines through A are λ_t and λ_∞; no incidence of A is removed; A gains K_a by (C6) and nothing else (A is not of the form $(x{:}s{:}1)$ in (C3)). P: (F2), no removal, gains M. X_t, $t \notin S$: (F2), no removal, gains M (as $t \notin T$), and no K_s since $(0{:}t{:}1) = (x{:}s{:}1)$ forces $s = t$. X_a: loses λ_a (the case $w = 0$), gains M ($a \notin T$) and K_a ((C3) with $x = 0$). X_t, $t \in T$: loses λ_t, gains $\lambda_{\sigma(t)}$ by (C1) and K_t; it is not on M. A point $(x{:}y{:}1)$ with $x \neq 0$: its plane lines are $[u{:}1{:}{-}(ux+y)]$ and $[1{:}0{:}{-}x]$ (solve $ux + vy + w = 0$ with $v = 1$, resp. $v = 0$); it is not on

vertex	neighbours in G	degree
A	λ_t ($t \in \mathbb{F}$), λ_∞, K_a	$q+2$
P	ℓ, λ_∞, [1:0:w] ($w \in \mathbb{F}^*$), M	$q+2$
X_t, $t \notin S$	ℓ, [w:1:$-t$] ($w \in \mathbb{F}$), M	$q+2$
X_a	ℓ, [w:1:$-a$] ($w \in \mathbb{F}^*$), M, K_a	$q+2$
X_t, $t \in T$	ℓ, [w:1:$-t$] ($w \in \mathbb{F}^*$), $\lambda_{\sigma(t)}$, K_t	$q+2$
(x:y:1), $x \neq 0$	[u:1:$-(ux+y)$] ($u \in \mathbb{F}$), [1:0:$-x$], and K_y if $y \in S$	$q+2$ if $y \in S$, else $q+1$
(x:1:0), $x \neq 0$	λ_∞, [1:$-x$:w] ($w \in \mathbb{F}$)	$q+1$
Q	λ_∞, λ_t ($t \in \mathbb{F} \setminus T$), K_s ($s \in S$)	$q+2$
H_a	[w:1:$-a$] ($w \in \mathbb{F}$), M, ℓ	$q+2$
H_t, $t \in T$	[w:1:$-t$] ($w \in \mathbb{F}$), M, $K_{\sigma(t)}$	$q+2$
ℓ	P, X_t ($t \in \mathbb{F}$), H_a	$q+2$
λ_∞	A, (x:1:0) ($x \in \mathbb{F}$), Q	$q+2$
λ_t, $t \notin S$	A, (x:t:1) ($x \in \mathbb{F}$), Q	$q+2$
λ_a	A, (x:a:1) ($x \in \mathbb{F}^*$), Q, H_a	$q+2$
λ_t, $t \in T$	A, (x:t:1) ($x \in \mathbb{F}^*$), $X_{\sigma^{-1}(t)}$, H_t	$q+2$
[u:1:w], $u \neq 0$	(x:$-ux-w$:1) ($x \in \mathbb{F}$), ($-1/u$:1:0), and H_{-w} if $-w \in S$	$q+2$ if $-w \in S$, else $q+1$
[1:0:w], $w \neq 0$	($-w$:y:1) ($y \in \mathbb{F}$), P	$q+1$
M	P, X_t ($t \in \mathbb{F} \setminus T$), H_s ($s \in S$)	$q+2$
K_a	(x:a:1) ($x \in \mathbb{F}$), Q, A	$q+2$
K_t, $t \in T$	(x:t:1) ($x \in \mathbb{F}$), Q, $H_{\sigma^{-1}(t)}$	$q+2$

TABLE 1. The neighbourhood of every vertex of $G = G(q; a, c)$. Here $T = \{0, 1, c\}$, $S = T \cup \{a\}$ and $\sigma = (0\,c\,1)$, so $\sigma^{-1}(0) = 1$, $\sigma^{-1}(1) = c$, $\sigma^{-1}(c) = 0$. The filler (C7) adds one more neighbour to exactly the vertices of degree $q+1$.

ℓ, so it loses nothing, and it gains K_s exactly when $y = s \in S$. A point (x:1:0) with $x \neq 0$: its lines are λ_∞ and [1:$-x$:w], and it gains nothing. Q: $1 + (q-3) + 4 = q+2$ lines, by (C5). H_s: q lines by (C2), M by (C4), and exactly one line from (C6): ℓ if $s = a$, $K_{\sigma(s)}$ if $s \in T$.

Lines. ℓ: no removal (the removed incidences are on lines λ_s), gains H_a. λ_∞: no removal, gains Q. λ_t, $t \notin S$: no change except Q. λ_a: loses X_a, gains Q ($a \notin T$) and H_a ((C2) with $w = 0$). λ_t, $t \in T$: loses X_t, gains $X_{\sigma^{-1}(t)}$ by (C1) and H_t; it does not contain Q. A line [u:1:w] with $u \neq 0$: its points are (x:$-ux-w$:1) and ($-1/u$:1:0); it is not a line through A, so it loses nothing; it gains H_s exactly when [u:1:w] = [u:1:$-s$], i.e. $s = -w \in S$. A line [1:0:w], $w \neq 0$: its points are ($-w$:y:1) and P, and it gains nothing (it is not of the form [w':1:$-s$]). M: $1 + (q-3) + 4 = q+2$ points. K_s: q points by (C3), Q, and exactly one point from (C6): A if $s = a$, $H_{\sigma^{-1}(s)}$ if $s \in T$.

Filler. The points of degree $q+1$ number $(q-1)(q-4) + (q-1) = (q-1)(q-3)$, and so do the lines. The map δ sends (x:y:1) $\mapsto$ [x:1:$-y$], a line [u:1:w] with $u = x \neq 0$ and $-w = y \notin S$, and (x:1:0) $\mapsto$ [x:0:-1], a line [1:0:w] with $w = -1/x \neq 0$. It is injective, so it is a bijection between the points and the lines of degree $q+1$. Since these points are not on ℓ, $p \notin \delta(p)$ (§1.1), so (C7) adds new edges forming a perfect matching between the vertices of degree $q+1$. □

1.5. **Diameter.**

Lemma 2. *Let $\mathcal{I}$ be a finite incidence structure in which every line has at least one point, any two points lie on a common line and any two lines meet in a common point. Then its incidence graph has diameter at most* 3*.*

Proof. Two points, or two lines, are at distance at most 2. Let p be a point and L a line with $p \notin L$. Take a point $p' \in L$; then $p' \neq p$. Take a line L' through p and p'; then $L' \neq L$, and p, L', p', L is a path. □

In the next two lemmas, *incident* means incident in $\mathcal{S}$. Recall that the only incidences of $\mathrm{PG}(2,q)$ missing from $\mathcal{S}$ are $X_s \in \lambda_s$, $s \in S$. So an incidence $u \in L$ of $\mathrm{PG}(2,q)$ holds in $\mathcal{S}$ whenever $u \notin \ell$ or L is not a line through A.

Lemma 3 (Points)**.** *Any two points of $\mathcal{S}$ lie on a common line of $\mathcal{S}$.*

Proof. Let $u \neq v$ be points.

(P1) Two points of $\mathrm{PG}(2,q)$. Let L be the line of $\mathrm{PG}(2,q)$ through u and v. If both incidences hold in $\mathcal{S}$, use L. Otherwise, say, $u = X_s$ and $L = \lambda_s$ with $s \in S$, and by (F1) $v = A$ or $v = (x{:}s{:}1)$ with $x \neq 0$. If $v = (x{:}s{:}1)$: K_s contains $u = (0{:}s{:}1)$ and v by (C3). If $v = A$ and $s = a$: K_a contains X_a ((C3), $x = 0$) and A ((C6)). If $v = A$ and $s \in T$: $\lambda_{\sigma(s)}$ contains X_s ((C1)) and A.

(P2) Q and a point u of $\mathrm{PG}(2,q)$. If $u \in \{A, P\}$ or $u = (x{:}1{:}0)$ ($x \neq 0$): λ_∞, by (F1) and (C5). If $u = X_t$, $t \notin S$: λ_t ($t \notin T$, and $X_t \in \lambda_t$ is kept). If $u = X_s$, $s \in S$: K_s ((C3) with $x = 0$, (C5)). If $u = (x{:}y{:}1)$, $x \neq 0$: $u \in \lambda_y$ by (F1); use λ_y if $y \notin T$, and K_y if $y \in T$.

(P3) H_s and a point u of $\mathrm{PG}(2,q)$. If $u \notin \ell$ (this includes A), write $u = (x{:}y{:}z)$ with $x \neq 0$: by (F3) u lies on $[(sz - y)/x{:}1{:}{-}s]$, which contains H_s by (C2). If $u = X_s$: the line $[1{:}1{:}{-}s]$ contains X_s, is not λ_s, and contains H_s. If $u = P$ or $u = X_t$ with $t \notin T$, $t \neq s$: M ((C4)). If $u = X_t$ with $t \in T$, $t \neq s$, and $s = a$: ℓ ((C6)). If $u = X_t$ with $t \in T$, $t \neq s$, and $s \in T$: since σ is a 3-cycle on T, either $t = \sigma(s)$ or $t = \sigma^{-1}(s)$. If $t = \sigma(s)$, use $K_{\sigma(s)}$, which contains $X_{\sigma(s)}$ ((C3)) and H_s ((C6)). If $t = \sigma^{-1}(s)$, use $\lambda_s = [0{:}1{:}{-}s]$, which contains $X_{\sigma^{-1}(s)}$ ((C1)) and H_s ((C2) with $w = 0$).

(P4) Two new points. H_s and $H_{s'}$: M. Q and H_a: λ_a (Q by (C5) as $a \notin T$; H_a by (C2) with $w = 0$). Q and H_t, $t \in T$: $K_{\sigma(t)}$ ((C5), (C6)). □

Lemma 4 (Lines)**.** *Any two lines of $\mathcal{S}$ meet in a common point of $\mathcal{S}$.*

Proof. Let $L \neq N$ be lines.

(L1) Two lines of $\mathrm{PG}(2,q)$. Let p be their common point in $\mathrm{PG}(2,q)$. If both incidences hold in $\mathcal{S}$, use p. Otherwise, say, $p = X_s$ and $L = \lambda_s$ with $s \in S$, and by (F2) $N = \ell$ or $N = [w{:}1{:}{-}s]$ with $w \neq 0$. If $N = [w{:}1{:}{-}s]$: H_s lies on N and on $\lambda_s = [0{:}1{:}{-}s]$ by (C2). If $N = \ell$ and $s = a$: H_a lies on λ_a ((C2), $w = 0$) and on ℓ ((C6)). If $N = \ell$ and $s \in T$: $X_{\sigma^{-1}(s)}$ lies on ℓ and on λ_s ((C1)).

(L2) M and a line L of $\mathrm{PG}(2,q)$. If L passes through P, i.e. $L \in \{\ell, \lambda_\infty\}$ or $L = [1{:}0{:}w]$ ($w \neq 0$) by (F2): P. If $L = \lambda_t$, $t \notin S$: X_t ($t \notin T$, and $X_t \in \lambda_t$ is kept). If $L = \lambda_s$, $s \in S$: H_s ((C2) with $w = 0$, (C4)). If $L = [u{:}1{:}w]$ with $u \neq 0$: by (F4) L meets ℓ in X_r, $r = -w$, and this incidence is kept. If $r \notin T$, use $X_r \in M$. If $r \in T$, then $L = [u{:}1{:}{-}r]$ contains H_r by (C2), and $H_r \in M$.

(L3) K_s and a line L of $\mathrm{PG}(2,q)$. If L does not pass through A (that is, $L = \ell$, or $L = [u{:}v{:}w]$ with $u \neq 0$): L meets λ_s in a point $(x{:}s{:}1)$ (by (F1) for $L = \ell$, where the point is X_s; by (F4) otherwise), the incidence with L is kept, and the point is on K_s by (C3). If $L = \lambda_s$: the point $(1{:}s{:}1)$ (on λ_s, not on ℓ) and K_s. If $L = \lambda_\infty$ or $L = \lambda_t$ with $t \notin T$: Q ((C5)). If $L = \lambda_t$ with $t \in T$, $t \neq s$, and $s = a$: A ((C6)). If $L = \lambda_t$ with $t \in T$, $t \neq s$, and $s \in T$: if $t = \sigma(s)$, the point X_s lies on $\lambda_{\sigma(s)}$ ((C1)) and on K_s ((C3)); if $t = \sigma^{-1}(s)$, the point H_t lies on λ_t ((C2), $w = 0$) and on $K_{\sigma(t)} = K_s$ ((C6)).

(L4) Two new lines. K_s and $K_{s'}$: Q. M and K_a: X_a ((C4) as $a \notin T$; (C3) with $x = 0$). M and K_t, $t \in T$: $H_{\sigma^{-1}(t)}$ ((C4), (C6)). □

Theorem 5. *Let $q \geq 4$ be a prime power and a, c distinct elements of $\mathrm{GF}(q) \setminus \{0, 1\}$. The graph $G(q; a, c)$ is bipartite, has $2(q^2 + q + 6)$ vertices, maximum degree $q + 2$ and diameter 3. The graph $R(q; a, c)$ has the same properties and is $(q + 2)$-regular.*

Proof. The order and the degrees are Lemma 1. Every line of $\mathcal{S}$ has at least one point. By Lemmas 3 and 4 and Lemma 2, G has diameter at most 3; adding the edges (C7) cannot increase it. The diameter is at least 3 because the graphs are bipartite and not complete bipartite (for instance $A \notin \ell$). □

Remark 6 (What the proofs use). Only that $\mathbb{F}$ is a field and that $0, 1, a, c$ are distinct: the divisions in (F3), (F4) and Table 1 are by $x \neq 0$ or $u \neq 0$, and the filler uses $x^2 \neq 0$ for $x \neq 0$. No assumption on the characteristic is needed. The values $0, 1, c$ enter only through T and the 3-cycle σ, so the same proofs hold verbatim for any three distinct elements $t_0, t_1, t_2 \in \mathbb{F} \setminus \{a\}$ and $\sigma = (t_0\, t_2\, t_1)$. The point P plays the same role as every X_t with $t \notin S$; it is not a parameter.

Remark 7 (Duality). Write $\mathcal{S}_\sigma$ for the structure built with the 3-cycle σ on T. The correlation δ maps the points of $\mathcal{S}_\sigma$ to the lines of $\mathcal{S}_{\sigma^{-1}}$ and vice versa, preserving incidence, if we extend it by $\delta(H_s) = K_s$, $\delta(K_s) = H_s$, $\delta(Q) = M$, $\delta(M) = Q$. For instance, δ maps $X_t \in \lambda_{\sigma(t)}$ to $X_{\sigma(t)} \in \lambda_t$, which is the rotation of $\mathcal{S}_{\sigma^{-1}}$, and maps $(x{:}s{:}1) \in K_s$ to $H_s \in [x{:}1{:}{-}s]$. So Lemma 4 also follows from Lemma 3 applied to $\mathcal{S}_{\sigma^{-1}}$; we have written it out so that each case can be checked directly. For $\mathcal{A}$, δ is a correlation of $\mathcal{A}$ itself.

Remark 8 (Computer check). The program `coords.py` builds $\mathcal{A}$ and $\mathcal{S}$ only from this section, in three ways (from the steps (C1)–(C6), from the point rows of Table 1, and from its line rows), checks that the three incidence sets coincide, and checks the degrees, both covering properties and the exact diameter of G and R, for every prime power $q \leq 13$ and every admissible (a, c): $2, 6, 20, 30, 42, 72, 110$ choices for $q = 4, 5, 7, 8, 9, 11, 13$. All pass. The cores G are isomorphic (by `nauty`) to those of the earlier coordinate-free implementation for $q = 9, 11, 13$, and the graphs R for $q = 9, 11, 13$ pass three independent diameter checkers. It also builds $\mathcal{A}$ for every prime power $q \leq 13$ and every a, confirms the degree count of §1.2 (with $i \in \mathbb{F} \setminus \{0, 1, a\}$ the matching of [1] leaves four vertices of degree $q + 1$; with $i \in \mathbb{F}^*$ the graph is regular), and finds the graphs isomorphic to an independent implementation of [1]. It checks the duality of Remark 7 directly. Removing the rotation, or the incidence $H_1 \in K_0$, gives diameter 4 in every case tested ($q = 5, 9, 13$).

References

[1] G. Araujo-Pardo and N. López, On new record graphs close to bipartite Moore graphs, *Graphs Combin.* (2022). `arXiv:2005.02427`.